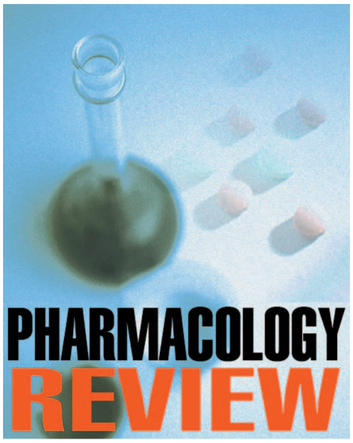

Clinical Pharmacology of Caffeine in the Newborn

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Abstract
Caffeine is used commonly in the neonatal intensive care unit to treat apnea of prematurity. This review describes the mechanism of action, pharmacokinetics, and therapeutic role of caffeine. Published data on its efficacy and safety document that caffeine can reduce the frequency of apneic episodes and appears safe in the short term.

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Drs Natarajan, Lulic-Botica, and Aranda did not disclose any financial relationships relevant to this article.

Introduction
Caffeine, a trimethylxanthine, is widely used as first-line drug therapy in apnea of prematurity. Its pharmacologic effect in apnea includes stimulation of the medullary respiratory center, increased sensitivity to carbon dioxide, and enhanced diaphragmatic contractility. (1) Other demonstrated effects include stimulation of the central nervous and cardiovascular systems, enhanced catecholamine secretion, an increase in the basal metabolic rate, and alteration in glucose homeostasis. (2) Cardiac output and heart rate are increased, while peripheral vascular resistance is lowered. Caffeine exerts most of its effects by blocking adenosine receptors A1 and A2a, increasing cyclic 3,5 adenosine monophosphate (AMP) by inhibition of phosphodiesterase, and translocating intracellular calcium. (3) Prostaglandin antagonist activity and, in animal models, upregulation of gamma-aminobutyric acid A receptor subunits in the brainstem also have been suggested. (4)(5)

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Pharmacokinetics
Caffeine is one of the drugs used in the neonatal intensive care unit for which extensive pharmacokinetic data are available, particularly in the preterm neonate. (6)(7) Biotransformation of caffeine occurs in the liver via microsomal cytochrome P450 mono-oxygenases (CYP1A2) and via the soluble enzyme xanthine oxidase. The predominant process of caffeine metabolism in the preterm infant is N7-demethylation, which matures at about 4 months of age. (8) N3- and N7-demethylation increase exponentially with postnatal age, regardless of birthweight or gestational age. (8)(9) The female neonate demonstrates a higher rate of caffeine metabolism than the male. (8) In animal experiments, caffeine causes a dose-dependent elevation of hepatic CYP1A1 and 2 activities in microsomal preparations and CYP1A1 and 2 mRNA concentrations over a dose regimen of 50 to 150 mg/kg per day for 3 days. (10) Through its inductive effects on N-demethylation and C-8 oxidation, it appears to increase its own metabolism. (11)

Pharmacokinetic studies in preterm neonates have established that the half-life of caffeine is prolonged to 102.9±17.9 hours and remains prolonged for as long as 38 weeks' gestation, which reflects a maturational deficit of hepatic biotransformation. (6) Plasma half-life and elimination rates reach adult levels at 3 to 4.5 months of age. (7) Caffeine half-life may be prolonged further in infants who have cholestatic jaundice and exclusively breastfed infants. (12) Caffeine is eliminated primarily by renal excretion in the first postnatal weeks. Population pharmacokinetic studies have established that postconceptional age and parenteral nutrition influence caffeine clearance in neonates; neonates of low gestation receiving parenteral nutrition, therefore, may need closer monitoring of caffeine concentrations. (13)

Therapeutic Drug Monitoring

Therapeutic concentrations of caffeine vary widely from 8 to 40 mg/L, with some overlap with subtherapeutic concentrations. (14) The recommended dose regimen includes a loading dose of 20 to 40 mg/kg caffeine citrate, which corresponds to 10 to 20 mg/kg caffeine base, followed by a maintenance dose of 5 to 8 mg/kg per dose of caffeine citrate starting 24 hours later. (15) Standard doses of 5 mg/kg per day following a loading dose of 20 mg/kg should achieve therapeutic levels in more than 70% of neonates. (16) The drug has a wide therapeutic margin, with serious toxicity reported only at plasma concentrations higher than 40 to 50 mg/L, if at all. (15)

A higher dose regimen (loading dose of 50 mg/kg caffeine citrate administered as two separate doses 1 hour apart and maintenance dose of 12 mg/kg once daily to produce a desired plasma concentration of 30 mg/L [range, 26 to 40 mg/L]) has been shown to reduce the number of apneic episodes by more than 50% within 24 hours of treatment compared with a one-third reduction with the standard dose. (17) A rapid improvement in apnea within 8 hours also was apparent in the infants treated with the higher dose regimen without any adverse effects. (17) In this particular study, 69% of infants receiving standard doses achieved plasma concentrations between 13 and 20 mg/L, and 73% of the higher dose group achieved concentrations in the range of 26 to 40 mg/L.

Another population pharmacokinetic study compared daily doses of caffeine citrate of 30, 15, and 3 mg/kg for 7 days following loading doses of 60, 30, and 6 mg/kg in 119 preterm infants who had apnea. (18) Mean serum concentrations after the final dose were 69, 35.8, and 7.4 mg/L in the treatment groups, and no adverse effects were reported. A dose of 20 mg/kg caffeine citrate resulted in reduced extubation failure compared with a dose of 5 mg/kg in another study without any adverse effects during a 12-month follow-up period. (19)

In a single report of a toxic overdose of 160 mg/kg in a 31-week gestation 1,860-g infant, the serum concentration was 217.5 mg/L 36.5 hours after dosing. Toxic manifestations included hypertonia, sweating, tachycardia, cardiac failure, pulmonary edema, metabolic acidosis, hyperglycemia, and creatine kinase elevation. (20) Monitoring of plasma concentrations generally is recommended if there is lack of clinical response or suspected toxicity. (21) Periodic routine drug monitoring also is performed at some centers, although the value of this practice is uncertain.

Drug Interactions

Because cytochrome P450 1A2 (CYP1A2) is the major enzyme involved in its metabolism, caffeine has the potential to interact with drugs that are substrates for CYP1A2. Cimetidine and ketoconazole can inhibit caffeine metabolism, necessitating lower doses of caffeine. Phenytoin and phenobarbital can increase caffeine elimination, possibly requiring higher doses of caffeine. (22) In adults, estrogen-containing contraceptives can decrease caffeine clearance by up to 65%, resulting in an extended elimination half-life. (22) Caffeine antagonizes the effects of adenosine, with a decreased therapeutic effect at standard doses.

Comparison With Theophylline

Although very similar in its actions to theophylline, caffeine has several advantages and has become the preferred methylxanthine in the treatment of apnea. Its toxicity is lower and half-life is longer, and there is less need for therapeutic drug monitoring (Table). A systematic review of three trials comparing theophylline and caffeine citrate at loading doses of 20 to 25 mg/kg, with maintenance doses of 2.5 to 6 mg/kg, revealed no difference in efficacy of reduction of apnea. (23) Adverse effects, such as tachycardia and feeding intolerance, were lower with caffeine. In neonates, caffeine is a biotransformation product of theophylline via methylation; in adults, the principal metabolic pathway for theophylline is demethylation and oxidation.

Efficacy

Apnea of Prematurity

Apnea is a commonly encountered problem in the neonatal intensive care unit, with a reported prevalence of 35% in infants born at less than 32 weeks' gestation and 84% in extremely low-birthweight neonates. (24) Caffeine currently is considered the pharmacologic treatment of choice for apnea of prematurity. The Cochrane review of five trials involving 192 preterm infants who had apnea indicated that methylxanthine

Table. **Comparison of Caffeine and Theophylline**

	Caffeine	Theophylline
Loading dose (LD)	20 to 40 mg/kg per dose IV/PO	4 to 8 mg/kg per dose IV
Maintenance dose (MD)	5 to 8 mg/kg per dose IV/PO caffeine citrate	1.5 to 3 mg/kg per dose IV
Plasma half-life (h)	40 to 230 (mean, 103)	12 to 64 (mean, 30)
Therapeutic level (mcg/mL)	5 to 25	7 to 12
Toxic level (mcg/mL)	>40 to 50	>20
Signs of toxicity	Sinus tachycardia, hypertonia, sweating, cardiac failure, pulmonary edema, metabolic disturbances	Sinus tachycardia, agitation, electrolyte abnormalities, significant diuresis, gastrointestinal bleeding, ventricular tachycardia, seizure
Cerebrospinal fluid (CSF) distribution	Similar to plasma concentrations (reported correlation coefficients of 0.77)	Crosses into the CSF (reported correlation coefficients of 0.54)
Volume of distribution (L/kg)	0.8 to 0.9	0.45
Metabolism	Excreted unchanged or CYP P450 (CYP1A2) liver-methyltransferase pathway	Excreted unchanged or CY P450 (CYP1A2) metabolism
Interconversion between two	3% to 8% converted to theophylline via CYP1A2	25% converted to caffeine via methylation
Routine measurement of blood concentrations	Not required	Required
Clearance (L/kg per hour)	0.002 to 0.017	0.02 to 0.05
Elimination	Neonates younger than 1 month of age excrete 86% unchanged in urine; first-order elimination	Neonates excrete approximately 50% of the dose unchanged in urine; first-order kinetics; at high concentrations (>20 mg/L), the drug elimination mechanism becomes saturated, resulting in concentration-dependent elimination (zero-order kinetics)

therapy is effective in reducing the number of apneic spells and the use of mechanical ventilation in the 2 to 7 days after starting treatment. (25) Caffeine, which was specifically evaluated in two trials involving 103 preterm neonates, had a relative risk of treatment failure of 0.46 (95% confidence interval [CI], 0.27 to 0.78). (26)(27) When used as prophylaxis to prevent apnea in two placebo-controlled studies involving 104 preterm infants, no differences were apparent between the groups in the proportion of infants who had apnea, bradycardia, or hypoxemic episodes or the use of positive-pressure ventilation. (28)(29) The sample sizes in the studies were small, however, and caffeine was used for a total duration of 96 hours in one study.

Postoperative Apnea

The risk of postoperative apnea following general anesthesia, which is predominantly central, persists for about 60 weeks after conception. (30) Three trials involving 78 infants at gestational ages of 40 to 44 weeks found a significantly lower relative risk of 0.09 (95% CI, 0.02 to 0.34) when caffeine citrate was used at a dose of 5 to 10 mg/kg during or after induction of anesthesia. (31) Although probably efficacious, caffeine is not used routinely as prophylaxis to prevent postoperative apnea due to concerns of possible adverse effects and lack of long-term data.

Extubation

The use of methylxanthines as respiratory stimulants when weaning preterm infants from mechanical ventilation appears to reduce the failure of extubation, although controversy persists about the dosing regimen, duration, and long-term effects of use in this context. An overall analysis of six published trials showed a decreased relative risk of 0.47 (95% CI, 0.32 to 0.70) and an absolute reduction of 27% in the incidence of failed extubation. (32) One of these trials suggested a preferential beneficial effect in infants born at less than 1 kg birthweight and extubated within the first week after birth. (33) Only two of the trials evaluated caffeine. More recently, a randomized, double-blind clinical trial of three dosing regimens of caffeine citrate (3, 15, and 30 mg/kg) for periextubation management of 127 ventilated preterm infants of less than 32 weeks' gestation showed no statistically significant difference in the incidence of extubation failure among the dosing groups. (34) However, the infants in the two higher dosing groups had statistically significant less documented apnea in the immediate periextubation period. A larger trial involving 234 neonates born at less than 30 weeks' gestation reported a significant reduction in failure to extubate among infants receiving 20 mg/kg per day caffeine citrate compared with 5 mg/kg per day (15% versus 29.8%; relative risk, 0.51; 95% CI, 0.31 to 0.85). (19) A significant difference in the duration of mechanical ventilation was observed in infants of less than 28 weeks' gestation who received the higher dose. No differences were noted in short-term adverse effects or at 12 months follow-up.

Other Effects

Lung Function

Caffeine increases the central respiratory drive, thereby improving oxygenation and ventilation and decreasing hypoxic episodes. In the large randomized, controlled Caffeine for Apnea of Prematurity (CAP) trial, the rates of bronchopulmonary dysplasia (BPD), defined as an oxygen need at 36 weeks postconceptional age, were 36.3% in the caffeine group and 46.9% in the placebo group, a statistically significant difference. (35) This promising effect on the incidence of BPD has been attributed to the diuretic, respiratory stimulant, and anti-inflammatory effects of caffeine. (35) Previously, other smaller studies showed a decrease in airway resistance and improved lung mechanics within 1 hour of caffeine therapy in infants who had BPD. (36) An improvement in respiratory system compliance after caffeine administration also has been reported in preterm infants who had resolving respiratory distress syndrome. (37) The clinical effect has been validated in an immature baboon model treated with surfactant, where early caffeine treatment was associated with better lung function, higher compliance, and significant decreases in ventilator support. (38) In a rat pup model that received neonatal caffeine, there was a 22% higher minute ventilation response to hypercapnia in males in the juvenile stage, which persisted until adulthood. (39) This reported long-term effect on respiratory control was speculated to be due to a persistent change in adenosinergic neurotransmission.

Patent Ductus Arteriosus

The CAP trial involving infants weighing less than 1,250 g at birth showed a statistically significant decrease in the incidence of patent ductus arteriosus (30% versus 40%) and in the rates of surgical ligation (4.5% versus 12.6%) in the group treated with caffeine citrate (loading dose of 20 mg/kg followed by 5 to 10 mg/kg maintenance). (35) This effect needs to be clarified in further studies evaluating this specific outcome. However, caffeine is known to have prostaglandin antagonistic activity, an effect that is demonstrable at the concentrations achieved in human plasma. (4) In addition, it is a diuretic and a vasoconstrictor, related to adenosine antagonism, both possible mechanisms of a real effect on the ductus.

Cardiac Effects

Caffeine has a positive inotropic and chronotropic effect on the heart. Tachycardia is a well-known adverse effect of methylxanthine use. (2) Heart rate variability also has been noted in infants after receiving methylxanthine, with the effect being more pronounced in the sickest infants. (40)

Central Nervous System Effects

Caffeine is a generalized central nervous system excitant, causing increased transmission of impulses across neurons and synapses and stimulation of the motor end-plate. The acute neurologic effects of caffeine include jitteriness, tremor, hypertonia, and rhabdomyolysis, although these occur at higher plasma concentrations. (2) The effect of caffeine on cerebral blood flow has been a focus of investigation, primarily because of the concern of hemorrhage or periventricular leukomalacia in preterm neonates. A significant decrease in blood flow velocity in the internal carotid artery (22%) and anterior cerebral artery (14%) was observed in 16 preterm neonates following an oral "pure" caffeine dose of 25 mg/kg without a concomitant decrease in cardiac output. (41) Others have shown no effect on cerebral hemodynamics at lower loading doses and at a maintenance dose of 2.5 mg/kg caffeine *base*. (42)(43)(44)

Oxygen Consumption, Weight Gain, and Growth

An increase in oxygen consumption and energy expenditure has been reported in preterm neonates after 48 hours of caffeine therapy, which persisted through 4 weeks. (45) Caffeine-treated infants in the study required lower environmental temperatures to maintain normothermia. (45) The CAP trial validated a reduced weight gain in the caffeine cohort, with the greatest difference noted after 2 weeks (mean difference −23 g), (35) an effect known previously in adults and animals. (46)(47) Long-term effects on growth are unknown.

Renal Function

The diuretic effect of caffeine, following an increase in renal blood flow and glomerular filtration, is well known in adults and experimental animals. (2) Caffeine has no effect on serum sodium, potassium, calcium, and phosphorus concentrations; urinary calcium excretion increases and serum creatinine decreases significantly in preterm neonates. (48) At the clinically used doses, however, effects on renal function in neonates are minimal. (2)

Glucose Homeostasis

In the neonate, caffeine, like theophylline, may increase blood glucose concentrations acutely due to increased glycogenolysis, although this is seen infrequently in the usual clinical situation.

Gastrointestinal Effects

Caffeine increases gastric secretion and reduces lower gastroesophageal sphincter pressure. An increased duration of acid gastroesophageal reflux has been reported on esophageal pH monitoring in neonates receiving caffeine treatment during the late postprandial time (more than 2 h after the beginning of a meal). (49) Other studies have reported a similar increase in reflux time in about 50% of infants treated with caffeine, which resolved 2 weeks after stopping therapy and improved with cisapride. (50)(51) The increase in episodes of reflux was independent of plasma xanthine concentrations or drug efficacy. (52) The mechanism of lower esophageal sphincter relaxation appears to be via an enhancement of cyclic AMP levels. (53)

Safety

Fetuses and newborns are exposed to caffeine via maternal intake of caffeine-containing foods and beverages. This widespread and extensive exposure to caffeine must be considered in the evaluation of the long-term effects of caffeine in the newborn and young infants.

The CAP trial demonstrated no apparent short-term toxicity related to caffeine, although long-term neurodevelopmental outcomes are awaited. In particular, the incidence of necrotizing enterocolitis did not differ between the caffeine and placebo groups. (35) A previous smaller trial involving 85 neonates had raised concerns about a possible association with necrotizing enterocolitis, as had data indicating a reduction in mesenteric blood flow velocities following caffeine administration. (26)(41)(54)

The rates of ultrasonographic signs of brain injury did not differ between the caffeine-treated and placebo groups in the CAP trial. (35) Currently, the data on long-term effects in preterm neonates are scarce and inconclusive. A study involving 73 very low-birthweight neonates, some of whom also had germinal matrix or intraventricular hemorrhage, demonstrated no adverse effects of neonatal caffeine use on the 18-month Bayley mental development score. (55) In fact, infants who had a history of neonatal methylxanthine therapy scored better at 18 months than infants not treated, regardless of hemorrhage status, although the sample size was small. Gunn and associates (56) had earlier reported no differences in growth and development at 12 months of age in 21 very low-birthweight infants treated with methylxanthines compared with 21 untreated infants.

Adenosine A1 receptors are widely distributed through the brain and are highly expressed in the hippocampus during development. (57) Adenosine is neuroprotective during ischemia by reducing the release of glutamate. There have been concerns that caffeine may worsen certain forms of ischemic brain injury through its antagonistic effects on adenosine and may interfere with neural cell proliferation, axonal growth, learning, and memory during development. (57)(58) In experimental studies, caffeine induced neuronal death in neonatal rat brain and cortical cell cultures. (59) In neonatal mice, therapeutic doses of aminophylline decreased the rate of anoxic survival in vivo. (60) In rat pups treated with daily caffeine, impaired motor skills and changes in locomotor activity were demonstrated. The effects were dependent on the developmental stages at which caffeine was administered and the age at which tests were performed. (61) Impairment of motor skills, although transient, appeared early after treatment. (61) In a newborn mouse model, however, caffeine exposure did not worsen excitotoxic lesions of the periventricular white matter mimicking human periventricular leukomalacia. (57)

Conclusions

Caffeine is a widely used drug in preterm neonates who have apnea. Several clinical trials have demonstrated its efficacy in reducing the frequency of apneic episodes and need for mechanical ventilation. There has been a resurgence of interest in the drug, with a large randomized trial showing a decrease in the incidence of BPD. Caffeine, at least in the short term, appears safe at the currently used doses. Further long-term studies on neurodevelopmental outcome are awaited.

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NeoReviews Quiz

8. The pharmacologic effects of caffeine in the treatment of apnea of prematurity include stimulation of the medullary respiratory center, increased sensitivity to carbon dioxide, and enhanced diaphragmatic contractility. Of the following, caffeine exerts *most* of its effects by:

   A. Antagonism of prostaglandin activity.
   B. Blockage of adenosine receptors.
   C. Enhancement of catecholamine secretion.
   D. Stimulation of phosphodiesterase.
   E. Upregulation of gamma-amino-butyric acid receptors.

9. Caffeine administration in preterm infants warrants careful consideration of its dosage because of the interactions between caffeine and several other drugs. Of the following, the drug *most* likely to increase caffeine elimination, thereby warranting a higher dose of caffeine, is:

   A. Cimetidine.
   B. Estrogen.
   C. Indomethacin.
   D. Ketoconazole.
   E. Phenobarbital.

10. A 12-day-old neonate, whose birthweight was 760 g and estimated gestational age at birth was 26 weeks, has recurrent episodes of apnea, bradycardia, and cyanosis. Caffeine is administered intravenously with a loading dose of 20 mg/kg, followed by a maintenance dose of 5 mg/kg per day. Of the following, the *most* likely adverse effect of caffeine at this dose and assuming plasma concentrations of caffeine in the therapeutic range in this infant is:

    A. Altered renal function.
    B. Decreased energy expenditure.
    C. Increased blood glucose concentration.
    D. Opening of the ductus arteriosus.
    E. Relaxation of the lower esophageal sphincter.